

Accuracy vs. Interpretability in Designer-Level Fashion Classification

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Abstract

Designer-level fashion classification presents a challenging fine-grained visual recognition problem, where subtle garment attributes coexist with strong contextual cues such as runway layout, lighting, and venue design. In this work, we study how model capacity and training strategy influence both classification performance and interpretability on the iDesigner FGVC dataset. We evaluate standard ResNet architectures alongside two lightweight custom models, FashionNet-Small and FashionNet-Medium, under three training regimes: unmasked images, background-subtracted images using person segmentation, and aggressive augmentation on unmasked data. While models trained on unmasked images achieve high accuracy, Grad-CAM analysis reveals heavy reliance on background shortcuts. Background removal via segmentation substantially reduces this reliance and encourages garment-centric feature learning, but introduces significant performance degradation for lower-capacity architectures. Augmentation mitigates some shortcut learning while preserving accuracy, though residual background dependence remains. Through qualitative analysis using Grad-CAM and nearest-neighbor retrieval in feature space, we show that masking plays a critical role in shaping semantically meaningful, garment-focused representations. Our results highlight a fundamental trade-off between accuracy and interpretability in fashion recognition and emphasize the importance of architectural capacity when background cues are removed.

I. INTRODUCTION

The fashion industry has increasingly become a compelling testbed for computer vision, offering imagery that is both structured and abstract. Fashion imagery contains semantically meaningful components such as garments, accessories, textures, cuts, and silhouettes, yet these elements appear within scenes filled with variation in pose, lighting, styling, and presentation. This combination of consistency and diversity makes fashion a uniquely challenging domain in which to segment, classify, and interpret visual features. In particular, understanding how different garments, patterns, or aesthetic conventions relate to one another requires models capable of reasoning over subtle, fine-grained visual cues.

In this project, our goal was to study how model complexity influences both performance and interpretability on a fashion-designer classification task. To guide our investigation, we focused on three core questions:

- How complex does a model need to be to achieve strong classification accuracy?
- How complex does it need to be to extract meaningful and interpretable features of the dataset?
- How can the features learned by a model be used to analyze fashion itself?

To explore these questions, we trained several standard convolutional neural networks, including multiple ResNet variants, and compared them against a lightweight architecture of our own design. A central goal of this project was to understand what the models were actually learning: whether predictions were driven by garment-level signals such as shape, texture, and color, or by persistent dataset correlations like runway backgrounds, lighting, and venue cues.

This distinction became critical in the final stage of the project. Although early experiments achieved high validation accuracy, analysis of learned representations revealed that many models—particularly larger ones—relied heavily on background cues rather than the garments them-

selves. To address this, we introduced a preprocessing step using a DeepLab segmentation model trained on COCO to identify people and remove background pixels. The resulting background-subtracted dataset enabled a more faithful evaluation of each model’s ability to reason about fashion features instead of spurious contextual correlations. This shift ultimately shaped the remainder of our analysis, allowing us to reexamine the relationship between model capacity and interpretability under conditions that better isolate garment-level visual signals.

II. DATASET

The dataset used in this project originates from the iDesigner Fine-Grained Visual Categorization (FGVC) competition, part of CVPR 2019. It contains 50,000 high-resolution runway images collected across numerous fashion shows, with each image labeled by the designer of the garment worn by the model. Designers span a diverse range of major fashion houses including: Alexander McQueen, Gucci, Prada, Chanel, and Ralph Lauren making the classification task both visually rich and fine-grained. Figure I shows several representative examples from the original dataset.



FIGURE I: THREE EXAMPLE RUNWAY IMAGES FROM THE IDESIGNER DATASET.

Initially, we used the standard cropped version of the dataset, where images center the model while preserving much of the runway and background context. However, as

we previously mentioned, we discovered that many models were relying heavily on background cues (e.g., runway textures, lighting gradients, venue architecture) rather than the garments themselves. This finding motivated the construction of an augmented, background-subtracted version of the dataset to ensure that classification performance reflected genuine garment-level understanding.

To accomplish this, we applied a DeepLab segmentation model (trained on the COCO dataset) to generate a binary person mask for each image. The predicted mask isolates the human figure, allowing us to remove the background and retain only the model and garments. This procedure preserves the semantic content of interest while eliminating the contextual biases that models were inadvertently exploiting. Figure II illustrates this process: the original image, the extracted person mask, and the resulting background-removed rendering.

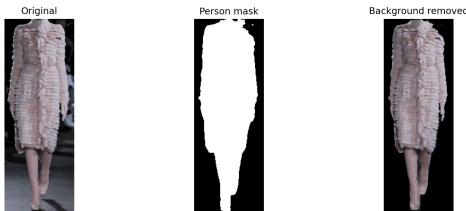


FIGURE II: EXAMPLE OF THE DEEPLAB SEGMENTATION PIPELINE APPLIED TO OUR DATASET. THE MODEL FIRST EXTRACTS A FOREGROUND PERSON MASK, WHICH IS THEN USED TO REMOVE ALL BACKGROUND PIXELS, LEAVING ONLY THE MODEL AND GARMENT.

This segmented dataset played an essential role in our final analysis, as it provides a cleaner foundation for evaluating the relationship between model complexity, interpretability, and true visual understanding of fashion features.

III. RELATED WORKS

Recent advances in computer vision for fashion analysis have been driven largely by progress in fine-grained visual categorization (FGVC), attribute recognition, and large-scale fashion datasets. Early efforts such as DeepFashion *et al.* [12] and DeepFashion2 *et al.* [5] established benchmarks for clothing classification, attribute prediction, and landmark localization, highlighting the importance of handling pose variation, subtle texture cues, and fine structural details. Jia *et al.* [9] introduced **Fashionpedia**, an ontology-driven dataset combining segmentation and fine-grained attribute localization, while Guo *et al.* [6] presented the **iMaterialist Fashion Attribute Dataset**, demonstrating how attribute-level annotations can enhance transfer learning within fashion tasks. Broader survey works such as Cheng *et al.* [1] further underscore the growing intersection between fashion and computer vision, outlining challenges in dataset diversity, intra-class variation, and interpretability.

Beyond dataset creation, models tailored for fine-grained fashion understanding have explored attribute

extraction, style recognition, and cross-domain retrieval. More recently, Parekh *et al.* [13] and Han *et al.* [7] examined multimodal pretraining and hierarchical attribute reasoning to better capture detailed visual semantics in apparel imagery. Additional work on fashion style classification, such as **FashionStyle14** benchmark introduced by Moeko Takagi *et al.* in What Makes a Style: Experimental Analysis of Fashion Prediction [14], demonstrates that high-level aesthetic concepts can be learned directly from visual features, though these style categories remain broader and less fine-grained than designer-specific distinctions. Fashion retrieval works — including street-to-shop matching by Kiapour *et al.* [10] and cross-domain fashion image retrieval by Bojana Gajic *et al.* [3] — highlight the need for representations sensitive to fine-grained differences in textures, fabric, and other characteristics that are also essential for distinguishing among designers.

More broadly, the FGVC literature provides architectural insights for modeling fine-grained variation in visual categories. Bilinear CNNs [11], Recurrent Attention CNNs [2], Multi-attention CNNs [16], and more recently transformer based methods like transFG [8] demonstrate strategies for leveraging local discriminative features and global contextual relationships. Meta-learning-based FGCV methods, such as Zhang *et al.* [15], further aim to improve sample efficiency and feature discrimination under limited data — an especially relevant consideration for fine-grained recognition, where visually subtle distinctions matter. Multimodal techniques have also influenced fashion representation learning. Models such as FashionBERT [4] and other text-vision pre-trained systems show that integrating textual metadata can enrich visual embeddings and enhance performance in classification, retrieval, and compatibility prediction. However, these methods rely on paired text descriptions, whereas our work focuses exclusively on the visual modality.

Despite progress across these domains and the significant progress made in fine-grained fashion understanding, we find no prior research utilising our specific dataset or exploring designer-level identification as a classification task. Existing fashion datasets and models focus primarily on attribute recognition, category classification, landmark detection, or retrieval, rather than attributing to its designer. Likewise no public available benchmark exists for this task. This work therefore explores a novel and narrowly defined problem setting, establishing a baseline performance on a small, exploratory dataset for designer recognition.

IV. METHODS

This section outlines the modeling, training, and evaluation procedures used in our study. Following the structure of our experimental pipeline, we describe the model architectures, the segmentation and masking procedure introduced after our midterm analysis, the augmentation-heavy pipeline adopted during our final phase of experiments, and the training and evaluation methodology used

across all models. We evaluate three families of convolutional neural networks: (1) ResNet baselines, (2) our custom *FashionNet-Small*, and (3) our extended *FashionNet-Medium*. This design allows us to compare large, general-purpose architectures against lightweight interpretable models.

RN18	RN34	RN50	FN-S	FN-M
11.69M	21.80M	25.56M	1.95M	5.28M

TABLE I: NUMBER OF LEARNABLE PARAMETERS PER MODEL ARCHITECTURE.

1. ResNet Baselines

We use ResNet-18, ResNet-34, and ResNet-50 as reference architectures. Each variant is trained both from scratch and with ImageNet pretraining. These models served as strong baselines due to their hierarchical feature extraction capabilities and robustness across diverse vision tasks. Their performance on both the unmasked and masked datasets provided a valuable point of comparison when assessing the behavior of lightweight models.

2. FashionNet

FashionNet-Small is a 3-stage convolutional architecture designed to emphasize local textures, mid-level garment patterns, and coarse silhouette information. It has approximately 1M parameters which makes it considerably smaller than the ResNet models and was initially developed to provide interpretability through more localized activations. Despite its simplicity, FashionNet-Small performed competitively on the unmasked dataset, but, as our analysis shows, relied heavily on background context. Following the failure of FashionNet-Small on the masked dataset, we designed FashionNet-Medium, a widened and slightly deepened version targeting roughly 5M parameters. The architecture increases channel widths across all stages and expands the third stage to better capture global garment structure. Its purpose is twofold: (1) to retain some interpretability advantages over large residual networks, and (2) to test whether increased capacity enables robust learning on datasets where background cues are removed or diminished. Our midterm analysis revealed that models trained on the unmasked dataset exhibited strong dependence on runway backgrounds and other contextual signals. To reduce this unintended bias, we introduced a segmentation-based masking pipeline inspired by the motivations outlined in the Fashionpedia dataset, which emphasizes localized garment understanding. We employed a person-detection and segmentation model (DeepLabV3-Resnet-101) to isolate the figure and remove background pixels. The resulting masked dataset preserved the clothing silhouette and texture while eliminating designer-specific runway cues. This dataset enabled us to directly evaluate each model’s ability to perform designer classification based on garment features alone. Although segmentation proved effective for reducing background reliance, we also explored a complementary strategy that does not require dataset reconstruction: strong

data augmentation. After observing the instability of FashionNet-Small on the masked dataset, we reverted to the original unmasked dataset and applied extensive augmentations to decorrelate clothing labels from superficial background patterns. The augmentation pipeline incorporated:

- random cropping and resizing,
- heavy color jittering,
- random horizontal flips and perspective transforms,
- random erasing applied at multiple scales,
- stochastic background-destroying transformations (e.g., coarse noise masks).

These augmentations were deliberately chosen to encourage garment-centric learning and to weaken the spurious correlations identified during our earlier analyses.

A. Training Procedure

All models were trained using cross-entropy loss with either SGD or Adam optimizers. For the ResNet models, we trained for 40 epochs in most settings, unless otherwise stated. FashionNet-Small and FashionNet-Medium followed the same training schedule for comparability across architectures. Learning rate schedules (step decay or cosine annealing) were chosen according to validation performance and stability. All models were trained on an Nvidia 5070 Ti.

B. Evaluation and Interpretability

We evaluate models using top-1 validation accuracy, but accuracy alone does not reveal how visual information is used. To analyze model behavior, we primarily rely on Grad-CAM visualizations across unmasked, masked, and heavily augmented datasets, which effectively expose background reliance and architectural differences between FashionNet and ResNet models. Occlusion tests produced qualitatively similar findings but are omitted due to space constraints. We further assess representation quality through nearest-neighbor retrieval in feature space, examining how learned embeddings shift under background removal and augmentation and whether models group garments based on meaningful visual similarity rather than contextual shortcuts.

C. Experimental Phases

Our full methodological workflow consisted of three chronological phases:

- Phase I: Baseline Training on Unmasked Dataset.** Models exhibited strong performance but relied heavily on background cues.
- Phase II: Masked Dataset Experiments.** Background removal exposed severe weaknesses in FashionNet-Small, while ResNets remained robust.
- Phase III: Augmentation-Based Mitigation.** Heavy augmentations on the unmasked dataset

reduced background dependence and enabled FashionNet-Medium to perform substantially better than FashionNet-Small.

This progressive, diagnosis-driven methodology allowed us to iteratively refine both our models and our understanding of the dataset, ultimately enabling a more accurate analysis of the fashion classification problem.

V. ANALYSIS AND RESULTS

Beginning with the unmasked dataset, we evaluate baseline performance and examine model behavior through Grad-CAM. We then analyze how performance changes when background information is removed via masking. Finally, we study how aggressive data augmentation alters the learned representations and affects each model’s ability to focus on garment features rather than contextual shortcuts.

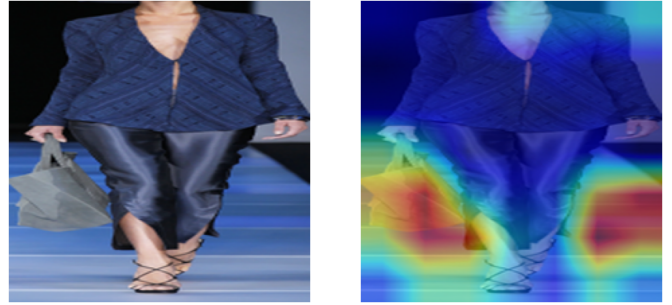
A. Phase I: Baseline Results on the Unmasked Dataset

Our initial experiments used the original, unmasked runway dataset. Table I summarizes validation accuracy on the unmasked dataset. Pretraining provides a large and consistent gain for ResNet models, with the best performance achieved by pretrained ResNet-50. Interestingly, when trained from scratch, FashionNet-Small and FashionNet-Medium outperform all scratch ResNet baselines, suggesting that the lightweight architectures may be particularly effective at exploiting consistent dataset cues. However, as shown by Grad-CAM in the following analysis, these gains do not necessarily reflect garment-centric reasoning, and much of the predictive signal appears to come from background correlations present in the runway setting.

Training	RN18	RN34	RN50	FN-S	FN-M
Scratch	79.40	79.88	81.56	86.64	89.02
Pretrained	92.78	93.14	94.14	–	–

TABLE I: VALIDATION ACCURACY (%) ON THE UNMASKED DATASET. FASHIONNET MODELS ARE TRAINED FROM SCRATCH ONLY.

The Grad-CAM visualizations (Figure IIIa–b) show that, despite the garment being clearly visible, the strongest activations are concentrated on the runway floor, surrounding lighting, and peripheral background regions rather than the clothing itself. In the illustrated example, large high-response regions appear on the floor and lateral runway edges, while the garment receives comparatively weaker attention. This behavior is consistent across architectures and model sizes, indicating that the dataset contains strong background–label correlations that models exploit to achieve high accuracy. As a result, performance on the unmasked dataset is partially driven by contextual shortcuts rather than genuine garment-level understanding.



(a) Original image

(b) Grad-CAM overlay

FIGURE III: EXAMPLE GRAD-CAM VISUALIZATION ON AN UNMASKED RUNWAY IMAGE. WHILE THE GARMENT IS VISIBLE, THE STRONGEST ACTIVATIONS FREQUENTLY OCCUR ON RUNWAY AND LIGHTING REGIONS, SUGGESTING RELIANCE ON CONTEXTUAL SHORTCUTS RATHER THAN GARMENT-ONLY CUES.

These observations motivated the next stage of experimentation, where background information is explicitly removed.

B. Phase II: Model Behavior on the Masked Dataset

To eliminate contextual cues, we constructed a masked dataset in which background pixels were removed using a person-segmentation model, forcing networks to rely on garment shape, texture, and local visual detail. Table II reports validation accuracy under this setting. Across all architectures, performance drops substantially relative to the unmasked dataset, confirming that background information previously provided a strong predictive signal. Pretrained ResNet models remain the most robust, retaining moderate accuracy despite the removal of contextual cues. In contrast, FashionNet-Small and FashionNet-Medium experience pronounced degradation, with performance approaching chance levels. This sharp decline suggests that the lightweight architectures lack sufficient representational capacity to reliably discriminate designers when restricted to garment-only information.

Training	RN18	RN34	RN50	FN-S	FN-M
Scratch	57.80	60.03	58.06	53.86	56.82
Pretrained	77.88	81.24	84.10	–	–

TABLE II: VALIDATION ACCURACY (%) ON THE MASKED DATASET. FASHIONNET MODELS ARE TRAINED FROM SCRATCH ONLY.

Grad-CAM provides insight beyond accuracy. When trained on the masked dataset, ResNet models produce more garment-centered activations, with salient regions concentrating on the clothing silhouette and mid-body structure rather than the runway floor or lighting. Figure IV shows an illustrative example using our best-performing ResNet-50 trained on masked images but evaluated on the original input. Despite the reintroduction of background at test time, gradients remain localized on the garment, suggesting that masking encourages features less dependent on contextual shortcuts. While Grad-CAM is not definitive evidence of causal reliance, this qualitative

shift aligns with the intended effect of background subtraction.

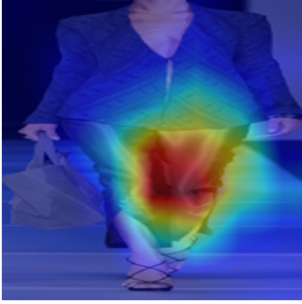


FIGURE IV: GRAD-CAM FOR A RESNET-50 TRAINED ON THE *masked* DATASET AND EVALUATED ON THE CORRESPONDING *unmasked* IMAGE. EVEN WHEN THE RUNWAY BACKGROUND IS PRESENT AT TEST TIME, ACTIVATIONS REMAIN CONCENTRATED ON THE GARMENT.

Quantitatively, Table II shows a substantial drop in validation accuracy across all models when training on masked data. This reflects both the loss of predictive context and a distribution shift introduced by masking. Background-subtracted images contain large near-zero regions, altering intensity statistics and reducing texture diversity. In practice, this made optimization more difficult, particularly for models trained from scratch. Normalization layers such as batch normalization are especially affected, as masked regions dominate feature statistics and reduce the effective signal-to-noise ratio in early layers. These factors help explain the sharp performance drop on the masked dataset and motivate Phase III, where we explore augmentation-based strategies that weaken background correlations without introducing extensive blacked-out regions.

C. Phase III: Heavy Augmentation on the Unmasked Dataset

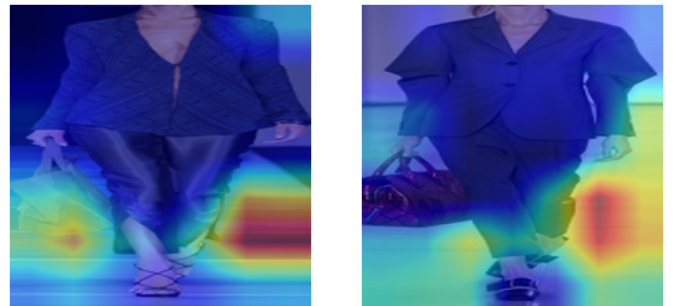
We explored a complementary strategy that preserves the original image structure while weakening contextual correlations by applying heavy data augmentation to the unmasked dataset. Models were trained with aggressive color jitter, geometric distortion, random erasure, and occlusion-like noise to disrupt background consistency without explicitly removing it.

Table III summarizes validation accuracy under this setting. Compared to the masked dataset, all models recover a substantial portion of their performance, indicating that augmentation mitigates some of the optimization difficulties introduced by background removal. Pretrained ResNet models remain strong performers, demonstrating robustness to aggressive appearance perturbations. FashionNet-Small improves relative to its masked performance but continues to lag behind larger architectures, while FashionNet-Medium exhibits more stable learning and consistently outperforms the smaller variant, suggesting that modest increases in capacity are sufficient to benefit from augmentation.

Training	RN18	RN34	RN50	FN-S	FN-M
Scratch	81.04	84.08	79.34	75.22	77.62
Pretrained	92.16	92.60	94.70	–	–

TABLE III: VALIDATION ACCURACY (%) ON THE UNMASKED DATASET WITH STRONG AUGMENTATION. FASHIONNET MODELS ARE TRAINED FROM SCRATCH ONLY.

Grad-CAM analysis reveals that augmentation partially shifts model attention toward garment regions, but important limitations remain (Figure V). Strong activations persist along the runway floor and surrounding scene, even when the garment is centrally visible. This indicates that while augmentation weakens some background correlations, it does not fully eliminate reliance on contextual cues. Instead, models appear to combine limited garment-level information with residual background signals that remain visually consistent across designers.



(a) Example 1

(b) Example 2

FIGURE V: GRAD-CAM VISUALIZATIONS FOR A RESNET-50 TRAINED WITH STRONG AUGMENTATION ON THE UNMASKED DATASET. THESE EXAMPLES SHOW THAT MODELS CONTINUE TO EXPLOIT PERSISTENT CONTEXTUAL CUES RATHER THAN RELYING EXCLUSIVELY ON GARMENT-LEVEL FEATURES.

Overall, these results refine our earlier conclusions. Both masking and augmentation reduce shortcut learning, but masking is substantially more effective at enforcing garment-centric representations. Models trained on masked data consistently exhibit more localized, garment-focused activations, indicating stronger alignment with fashion-oriented visual reasoning. However, this benefit comes at a clear cost in classification accuracy, particularly for lower-capacity architectures. Higher-capacity and pretrained models are better able to absorb this shift, suggesting that effective background removal must be paired with sufficient representational capacity to avoid severe performance degradation.

D. Feature-Space Behavior

To further probe what visual structure the model learned beyond classification accuracy, we performed a nearest-neighbor analysis in feature space using the FashionNet-Medium model trained on the masked dataset. Feature embeddings were extracted from the penultimate layer, normalized, and compared using cosine similarity. For each query image from the validation set, we retrieved

the top nearest neighbors from the training set and visualized both the ground-truth and predicted designer labels.

Although we are not fashion experts, the retrieved neighbors exhibit qualitatively meaningful similarities. Across multiple examples, nearest neighbors often share consistent color palettes, fabric textures, silhouettes, and overall aesthetic structure with the query image. In several cases, garments with similar layering, drape, or pattern geometry appear among the closest matches, even when the designer labels differ. This suggests that the model has learned a representation that organizes images according to visually salient garment attributes rather than relying solely on coarse scene context. Figure VI showcases some of these examples.

Importantly, this analysis uses the FashionNet-Medium model trained exclusively on masked images but evaluated here on the original unmasked dataset. Despite the reintroduction of background information at test time, the nearest-neighbor structure remains visually coherent and garment-focused. This provides further qualitative evidence that training with background removal encourages the model to encode intrinsic clothing features in its representation, even when background cues are later present. Taken together with the Grad-CAM results, these findings reinforce the conclusion that masking—while detrimental to raw classification accuracy—plays a critical role in shaping a more semantically meaningful feature space for fashion-oriented analysis.

E. Summary of Findings

In this work, we investigated how model capacity and training strategy influence both performance and interpretability in designer-level fashion classification. While standard training on unmasked runway images yielded high accuracy, qualitative analysis revealed heavy reliance on background context rather than garment features. Background removal via segmentation proved effective at encouraging garment-centric representations, as evidenced by Grad-CAM and nearest-neighbor analysis, but introduced substantial performance degradation for lower-capacity architectures. Larger and pretrained models were more robust to this shift, highlighting the importance of sufficient representational capacity when contextual cues are removed. Overall, our findings suggest that masking is

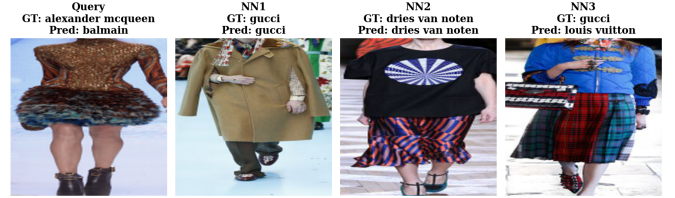
a powerful tool for promoting fashion-oriented visual reasoning on this dataset, but must be paired with appropriately expressive architectures to balance interpretability and accuracy.



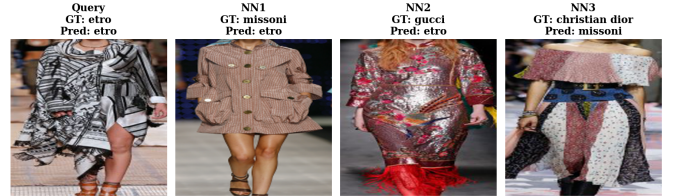
(a) Color-consistent neighbors



(b) Color-consistent neighbors



(c) Pattern and texture similarity



(d) Pattern and texture similarity

FIGURE VI: NEAREST-NEIGHBOR RETRIEVAL IN FEATURE SPACE USING FASHIONNET-MEDIUM TRAINED ON MASKED DATA. EACH ROW SHOWS A QUERY IMAGE AND ITS NEAREST NEIGHBORS FROM THE TRAINING SET.

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